Yuhan Chi

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Profile

Undergraduate at Fudan University, pursuing a double degree in Information and Computing Science and Artificial Intelligence. Research interests include mathematical interpretability for AI trust, verification and evaluation of LLM reasoning, and reinforcement learning for sequential decision-making.

Education

Fudan University, School of Mathematical Sciences Sep. 2025–Present
B.S. Candidate in Information and Computing Science & Artificial Intelligence (Double Degree).
GPA: 3.96/4.00 (most recent semester); rank: 10th (top 5%) in cohort.

Research Interests

- **Mathematical interpretability and AI trust:** grounding reliability claims in inspectable mathematical and statistical evidence.
- **Reasoning verification and evaluation:** studying when surface-level signals from LLM traces are diagnostic, misleading, or protocol-dependent.
- **Reinforcement learning and sequential decision-making:** using RL concepts to reason about uncertainty, feedback, and interaction with environments.

Selected Open-Source Research Projects

token-verification-mirage GitHub AI4Math Workshop
Solo-author project; poster, ICML 2026 Workshop on AI for Math (AI4Math).

- Led the full pipeline from dataset selection and model deployment to trace generation, evaluation design, analysis, figures, related-work organization, and writing.
- Audits how protocol choices such as pooling, in-sample scoring, and direction-agnostic AUROC affect apparent token-level verification performance.
- Shows that shallow token statistics can be useful diagnostics, but should not be treated as stable standalone verifiers without stronger controls.

code-not-text GitHub Demo
Solo project; cross-domain measurement study of cheap CoT-surface and token-trajectory features.

- Studies DeepSeek-R1-0528-Qwen3-8B across math, science, and coding tasks with problem-grouped evaluation.
- Reports strong math signal (AoA 0.958, AUROC@100% 0.982), partial science signal (AoA 0.799), and weak coding transfer (AoA 0.434).
- Tests coding robustness with an 83-scalar feature sweep, grouped ablations, a CoT-only judge, nonlinear MLPs, SSL pre-training, and token-level de-knotting.
- Frames the result as measurement non-invariance: the same feature family tracks convergence-like behavior in math but not executable correctness in coding.

TinyLoRA-GRPO-Coder GitHub
Solo project; small-parameter adaptation and RL training pipeline for code generation.

- Independent reimplementation and adaptation inspired by *Learning to Reason in 13 Parameters*.
- Moves a small-parameter adaptation and GRPO-style training pipeline toward verifiable competitive-programming code generation on Qwen2.5-Coder-3B.
- Uses compile-and-run rewards rather than static heuristics; built as a full research-system exercise covering data, training, reward design, evaluation, and validation.

microgpt.cpp GitHub
Solo project; minimal GPT implementation from first principles in C++.

- Implements core transformer components in C++ to make data flow, tensor operations, and model internals explicit and inspectable.

Academic Service

- Reviewer, ICML 2026 Workshop on AI for Math (AI4Math), 2026.

Skills

Programming	Python, C++, C, Java, Node.js, Lean 4
ML/AI	PyTorch, ML experimentation, LLM evaluation, reinforcement-learning basics
Tools	Git, GitHub, Linux, LaTeX, Markdown, VS Code
Other	Mathematical proof writing, technical writing, self-directed learning, competitive programming

Additional Information

- Maintains github-unflag-playbook-cn, a Chinese playbook documenting GitHub account flagging, recovery cases, and appeal workflows.
- Contributor to FDUGuideBook/nav-site, a Fudan community site.